Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026, Date: 24/02/2026

Mid-Semester Examination, Duration: 120 mins, Maximum Points: 36

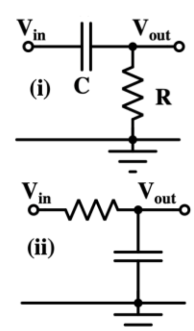
Instructions:

1. This assessment accounts for max. of 20 marks towards your total marks.
2. Show all your work clearly. Partial credit for correct reasoning.

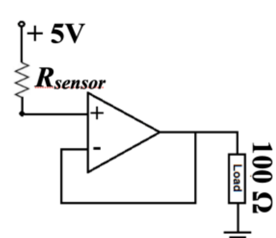
You want to become a part of the team designing electronics for a humanoid robot powered by a 24 V battery. ***Assume ideal op-amps (unless stated) and powered by ±15 V supplies. Assume ideal diodes unless specified.***

<u>Stage 1:</u> Before recruiting you on this project, you have been given a sheet to check your understanding and decide if you want to move to the next stage. You are asked to answer all questions and provide a one-sentence justification.

1. (1 point) In a purely DC circuit that has been turned on for a long time (steady-state), a capacitor acts as ____ circuit.
2. (1 point) At the instant an ideal inductor is connected to a DC source, what is its current at that instant (assuming zero initial current)? Justify in one sentence.
3. (1 point) A standard 50 Hz AC signal is passed through a Full-Wave Bridge Rectifier. What is the frequency of the resulting ripple voltage?
4. (1 point) State why we assume current flowing into the inverting (−) and non-inverting (+) input terminals of an ideal Op-Amp is nearly zero.
5. (1 point) An Op-Amp (Open loop gain = 100,000) with a grounded non-inverting input outputs −5 V. Calculate the exact voltage at its inverting input to prove it sits at virtual ground.
6. (1 point) An Op-Amp happens to have an open-loop gain of 100. If the input voltage at the inverting terminal is 1 V with grounded non-inverting terminal, what will be the output voltage?
7. (2 point) Consider a simple RC circuit as shown in the figure: (i) Output taken across the resistor, (ii) Output taken across the capacitor. In which configuration are low-frequency input signals passed while high-frequency signals are attenuated? Justify briefly.



8. (1 point) Build an OR-gate using two diodes.
9. (1 point) A sensor outputs exactly 5 V by itself, but the voltage drops to just 1 V the moment you connect a 100 Ω resistor to it. Why?
10. (1 points) The sensor in Q9 is connected to the 100 Ω load through an ideal unity-gain buffer as shown in the figure. Recalculate the voltage across the load. Justify your answer.



Stage 2:

Q11. *The robot's control unit must not start until the power rail stabilizes. We use an RC delay circuit to hold the boot signal.*

A delay circuit is formed by connecting a 12 V DC supply to a series resistor R = 100 kΩ and a capacitor C = 22 μF tied to ground. The capacitor is initially uncharged. The control unit measures the voltage across the capacitor and is programmed to turn ON exactly when the voltage reaches 9 V.

i. (1 point) Calculate the time constant of the charging circuit.

ii. (2 points) Calculate the exact time in seconds it takes for the capacitor to reach 9 V if the control unit draws zero current from the delay circuit.

iii. (4 points) On the other hand, if the robot's control unit acts as a 50 kΩ load across the capacitor, recalculate the circuit's time constant and maximum voltage to determine if it will ever reach the threshold to turn on.

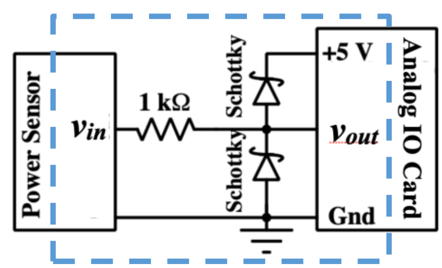
Q12. *The robot's arm uses a servo motor. To program the motor controller safely, you must analyze how the motor's coils behave during both DC diagnostic testing and live AC operation.* The robot servo motor is modeled as a series circuit with a coil resistance of 3 Ω and an inductance of 4 mH.

i. (3 points) During the DC test, the motor is connected to a steady 12 V DC source until it reaches steady-state, at which point the cable is abruptly pulled, forcing the current to zero in exactly 10 μs. Calculate the steady-state DC current, and the magnitude of the induced kickback voltage (across the inductor) during the disconnection.

ii. (3 points) If the motor is driven by its normal AC operating signal v(t) = 12 cos(1000t) V, calculate the peak AC current and determine the phase angle of the current relative to the voltage. Briefly comment on why the magnitude of peak AC current is different from the steady-state DC current calculated in part (i).

Q13. *Heavy factory motors create AC voltage spikes that must be clipped before they reach the delicate microchips.*

A input signal $v_{in}(t) = 20 \sin(\omega t)$ passes through a 1 kΩ resistor to an output node that is connected to two Schottky diodes (with forward drop $V_D = 0.2$ V) tied to +5 V and ground power rails as shown in the figure.



i. (5 points) Determine the maximum and minimum output voltages.

ii. (3 points) Calculate the peak current through the resistor, and briefly explain why this series resistor is essential for preventing destruction of the circuit?

Q14. *A sensor outputs a weak negative signal ranging from −0.2 V to 0 V. Your microcontroller needs this signal scaled to a standard 0 V to +10 V range, but it can be destroyed if it receives any negative voltage.*

Design an inverting Op-Amp circuit (R_{in} = 2 kΩ) to scale a -0.2 V sensor signal to a +10 V output, incorporating an ideal diode across the feedback resistor (R_f) with its anode at the inverting input to protect against positive input spikes.

i. (2 points) Calculate the required feedback resistor value (R_f).

ii. (2 points) Determine the voltage at the input of the microcontroller if the sensor fails and outputs a +2.0 V spike?